Khlong Sakae Bridge Barge Strike Inspection

Pa Sak Structural Diagnostics Co., Ltd. for Krungkao Crossing Co., Ltd. · Ref. PSD-IR-2026-031 · Issued 18 September 2026

1. Incident and scope

At about 04:40 on 14 September 2026, a loaded sand barge drifting downstream on the Pa Sak River struck the upstream nose of Pier P3 of the Khlong Sakae Bridge, the four-span toll bridge (KC-04) in Ayutthaya owned by Krungkao Crossing. Pier P3, its bearings and spans 2 and 3 were inspected on 15 September, and the P3 pile cap was surveyed underwater on 16 September.

2. Findings

- Pier P3 upstream nose: concrete spalled over an area of about 1.6 m by 0.9 m, with two main reinforcing bars exposed but not bent. Two diagonal cracks of 0.4 mm and 0.25 mm run from the spall.
- Pile cap: the steel fender frame on the upstream side is torn over a length of 3.2 m and two of its five timber rubbing strips are missing.
- Bearings at P3: the elastomeric bearing under girder G2 of span 3 has shifted 12 mm laterally; the other seven bearings at P3 are within the 5 mm tolerance.

The structure is rated **Condition Class C (serious defect, safe under restriction)**. It was Class B at the routine inspection of 22 February 2026.

3. Interim traffic restrictions

The bridge reopened at 16:00 on 17 September 2026 with the following restrictions:

- Gross vehicle weight limit: 18 tonnes (the normal limit is 25 tonnes). Buses on the Route 9 shuttle service are exempt up to 21 tonnes but must cross one at a time.
- Speed limit: 30 km/h on spans 2 and 3; 50 km/h elsewhere on the bridge.
- River traffic: no vessel may pass through the P3–P4 navigation opening. Vessels use the P2–P3 opening only, from 06:00 to 18:00.

4. Actions and responsibilities

1. Fit a temporary steel jacket to the P3 nose and seal the cracks by 2 October 2026. Responsible: Krungkao Crossing maintenance contractor, supervised by Wilaiwan Thongmuen, Asset Manager.
2. Reset the G2 bearing during a night closure (22:00 to 05:00). Responsible: Pa Sak Structural Diagnostics, led by Chatchai Ruangsrisuk, Lead Inspector.
3. Repeat the underwater scour survey by 30 October 2026. Responsible: dive team under Arthit Bunyasarn, Dive Supervisor.

Crack gauges are read every 3 days by the toll plaza duty engineer. The restrictions in section 3 can only be lifted by written approval of the Lead Inspector after actions 1 and 2 are complete.

Fictional document created for testing a question-answering system. All names, figures and organisations are invented.